

Manual

Personnel Scheduling Reports PEP-APP 8.2

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1 Personnel Scheduling Reports: Overview

Purpose

This function package contains personnel scheduling evaluation and planning functions.

Implementation notes

You use the function package if:

- you want to compare personnel requirements to available personnel capacities in graphic planning;
- you want to plan the assignment of employees to workplaces in HYDRA and display the resulting personnel schedule.

Integration

This function package requires the function package *Management Functions for Personnel Scheduling (PEP-VWF)*.

Features

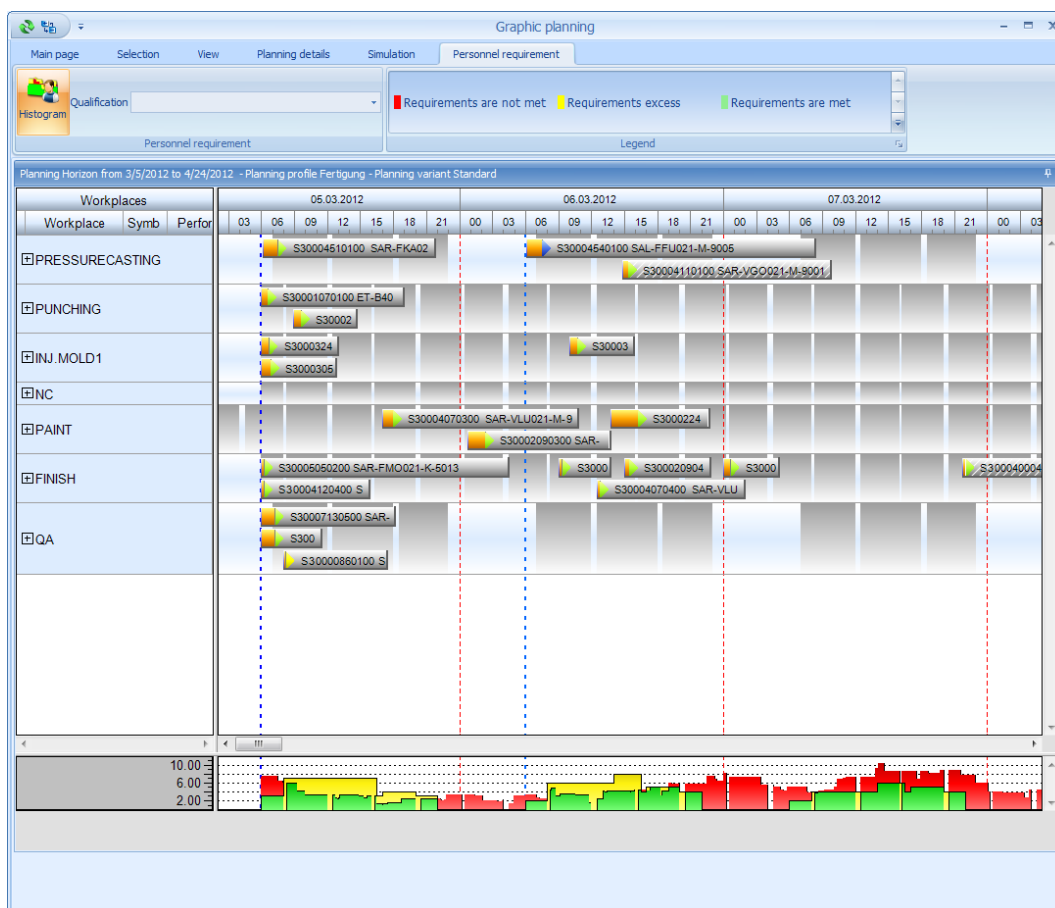
- Overview of workforce requirements in graphic planning
 - Graphical display of personnel requirements with variable time horizon and comparison of available employees with their qualifications
- Workplace assignment
 - Graphical display of workplaces with personnel requirements and available personnel capacities
 - Shift-related assignment of employees to workplaces
- Workplace schedule
 - Report showing the employees planned on workplaces
- Workforce requirements plan
 - Staff schedule in tabular format listing assigned workplaces

2 Graphic planning - personnel requirements

Overview

HYDRA menu	Production control → Preparations for production → Graphic planning
FEDRA menu	Detailed scheduling → Planning → Graphic planning
Tab	Personnel requirements
Function authorization	perov

You can use the function *Personnel requirement* of the graphic planning to show and compare the required and available personnel in a histogram and a table.



Requirements

If you want to show the planned personnel capacities, you have to make sure that the absence and shift planning are maintained properly in HYDRA. Usually, you can do this in the [labor time maintenance](#) application.

In addition, you have to define personnel requirements. To do so, you can choose from the following three options. Irrespective of how the personnel requirements are defined, the system only generates personnel requirements if operations are planned.

- [Workforce requirements of workplaces](#)
- Workforce requirements defined by the machine/operator relation (M/O relation for setup, M/O relation for production) of [operations](#)
- Workforce requirements defined by the [production resources and tools](#) (resource types *PRU* for setup and *PER* for production)

By default, the system only shows the personnel requirements if personnel is available. Use the following INI entry to ignore the personnel availability. If you use this INI entry, the system will show the personnel requirements even if no personnel is available. Note: If you enable this INI entry and personnel is available, the system will not show the available personnel.

INI entry:

- Name: HLS
- Section: SCHEDULING
- Key: IGNORE_PERSONNEL_AVAILABILITY
- Value: 1
- Active: Enable/check the "active" checkbox

Toolbar

The toolbar tab *Personnel requirement* provides the following functions:

Histogram

Use this button to show or hide the histogram of the required and available personnel.

Table view

Use this button to show or hide the table view of the required and available personnel.

Qualification

You can restrict the displayed personnel requirements to one or multiple qualifications. Displaying all qualifications shows whether enough personnel is available. If you restrict the displayed data to one qualification, you can check if enough employees with this qualification are available.

Legend

If activated, the histogram shows different colors depending on whether:

- personnel requirements are met
- personnel requirements are not met
- too many employees are planned

Color	Meaning
Red	Personnel requirements are not met.
Yellow	Personnel requirements are exceeded.
Green	Personnel requirements are met.

Detail application: *Histogram*

The histogram graphically displays the required and available personnel. The histogram uses the colors illustrated in the legend to highlight if personnel requirements are exceeded or not met. The Y axis shows the number of staff that is available or required over time.



Hold the left mouse button and increase the scale of the workforce requirements (Y axis) if you want to increase the dialog that shows the personnel requirements histogram.

Detail application: *Table view*

The table view compares the required and available personnel in table format. The application shows the totals of required and available personnel in hours for each day.

Example: a person has a planned working time of 8 hours a day. An OP that requires 1 person and takes 5 hours is planned during these 8 hours. Results for this day:

Available personnel: 8 hours

Required personnel: 5 hours

Difference: 3 hours.



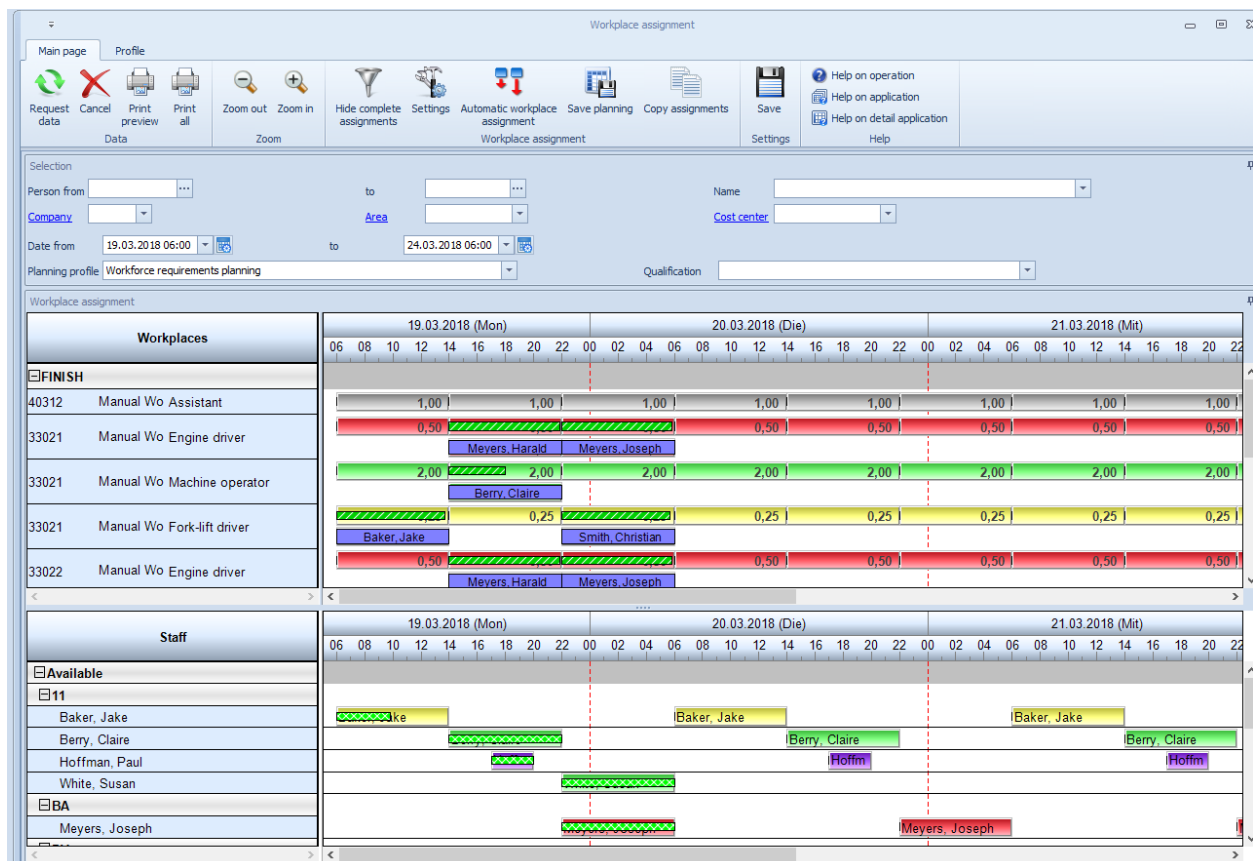
Personnel requirements are added up, if you plan multiple operations simultaneously for one workplace.

3 Workplace Assignment

Overview

HYDRA menu	Production control → Workforce requirements planning → Workplace assignment Human resources management → Workforce requirements planning → Workplace assignment
FEDRA menu	Advanced resource planning → Planning → Workplace assignment
Transaction code	wpas
Function authorization	wpas

Staff can be planned on workplaces using the workplace *Assignment function*. Planning can be performed manually as well as automatically.



Display

The center of the application shows the workplaces including the required qualifications grouped by workplace groups. The workforce requirements for the relevant qualification are shown in the bar for each shift. The bar length depends on the planned year model or on the year model of the [workplace](#). Only the workplaces for required workforce are displayed.



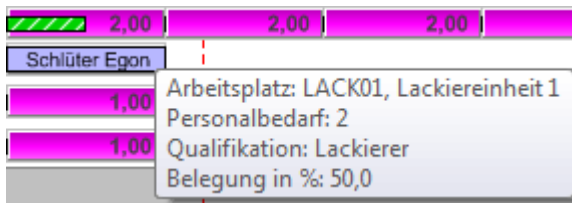
The "workplace assignment" does not allow for the breaks resulting from the employees' planned working time and that of workplaces. Instead, the workplace assignment plans entire shifts so employees and workplaces have continuous bars.



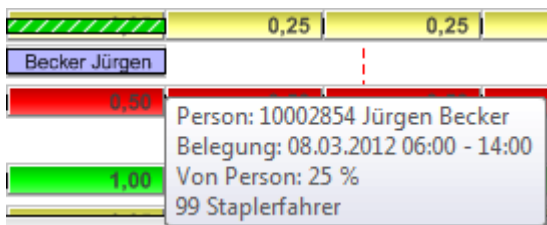
The format of the date values displayed on the Gantt chart depends on the format specified by the operating system. The client format is not relevant.

The lower section of the workplace assignment shows the available staff grouped by areas and the unavailable staff grouped by the abbreviations for absence reasons. The bar length depends on the standard working times (beginning of normal time and end of normal time) from the planned working time day type.

A green hatched bar shows the assignment rate in percent per shift at a workplace. The shift tooltip shows the percentage rate as figure. In addition, the planned staff is displayed:



The tooltip of an assigned person provides information about the person, their qualifications and assignment:



There are different ways of presenting for the assigned staff:

Presentation	Description
	The employee provides the required qualification
	The employee does not provide the required qualification
	Assigning the employee exceeds the workforce requirements for the workplace

Requirements

If you want to show the planned personnel capacities, you have to make sure that the absence and shift planning are maintained properly in HYDRA. You can do this in the [Personnel scheduling](#) application.

You also have to define personnel requirements. You can do this in three different ways:

- [Workforce requirements for workplaces](#)
- Workforce requirements defined by the machine/operator relation (M/O relation for setup, M/O relation for production) of [operations](#)
- Workforce requirements defined by the [production resources and tools](#) (resource types *PRU* for setup and *PER* for production)

You must plan operations in detail if the planned operations should be included in the workplace assignment. You can do this, for example, in "graphic planning". As an alternative, the ERP system might already have planned the operations in detail to send them to HYDRA.



Please note that "workplace assignment" and "graphic planning" should not be used simultaneously in production preparation. Inconsistencies might occur if personnel and capacities are planned at the same time, as the "workplace assignment" uses planning data of "graphic planning".

Selection criteria

The user selects personalized data. The application provides the following criteria for the user to select:

Qualification

The personnel requirements to be displayed and the available persons can be restricted to specific qualifications. This field does not act as a selection criterion when you request data, but filters the qualifications in the data displayed. This allows different views of the currently displayed planning without having to save the information and request the data again.

Planning profile

Using the planning profile, personnel assignment can be restricted to specific workplace groups and employees.



This field is only available if the additional function "enhanced selection and visualization" (PEP-ESV) is available (only applicable if HYDRA is used).

Toolbar



Zoom in,



Zoom out

These buttons reduce or increase the displayed workplaces and employees.



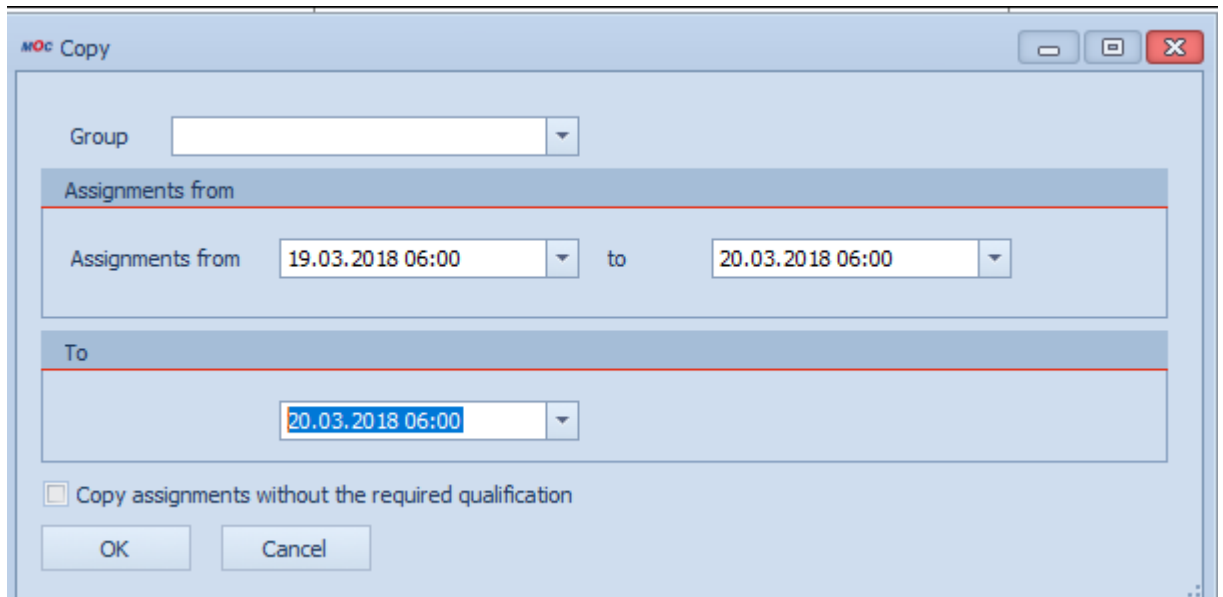
Hide complete assignments

Using this button, you can hide fully occupied workplaces and employees in order to get an overview of open requirements and available capacities. Workplaces and employees are considered being fully assigned, if at least 95% of the requirements and/or capacity are occupied.



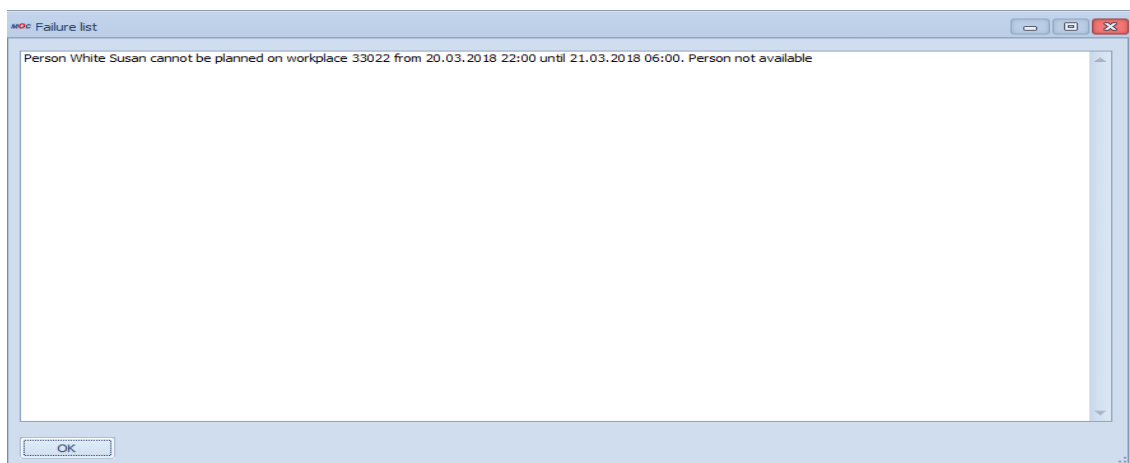
Copy assignments

Clicking this button opens a dialog where you can copy the assignment of one period (e.g. a day or week) to another period:

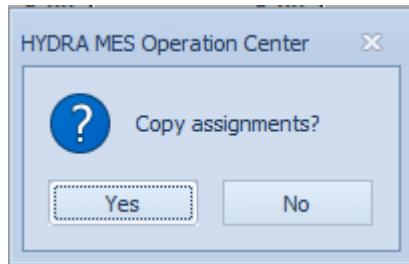


You can restrict the copy option to specific workplace groups. If the field "group" remains empty, you copy the assignments of all displayed workplaces.

An error list shows the assignments that could not be copied:



After closing the error list, the following dialog appears to decide if you want to copy the assignments despite the occurred errors:



All assignments are copied coinciding with the indicated period. The start period and the target period for copying must completely coincide with the period selected in the "workplace assignment". If this is not the case, copying is canceled issuing the error message "invalid period".



Lock all

When you attempt to plan an employee on a workplace, the system checks if another user is currently planning the workplace. If this is the case, the following error message occurs:

"The person ... is locked by user ..."

and/or

"The workplace ... is locked by user ..."

Using the button "lock all", you can lock all displayed employees and workplaces. Therefore, you can verify at the beginning of planning if all employees and workplaces to be planned are available.

If you locked employees and workplaces (with your user data), they are highlighted with green background color. A red background color indicates that employees and workplaces locked by another user.

Upon saving assignments, the following confirmation prompt appears:

"Do you want to release locked staff and workplaces?"

Here you can decide if you want to unlock employees and workplaces or if they should remain locked in order to continue planning.

Upon finishing the "workplace assignment", the system automatically releases the locked employees and workplaces.

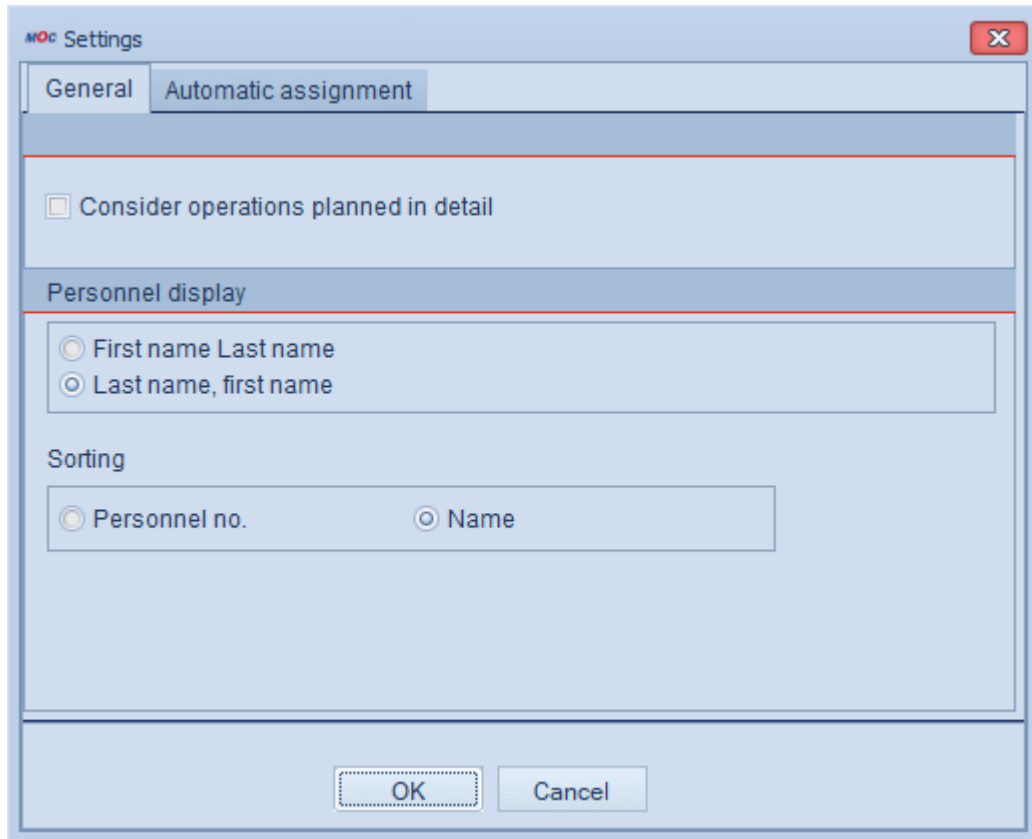
The following entry in the [INI configuration](#) disables the option of locking employees and workplaces as well as the check specifying if locks are available in the whole system:



Name:	PEP
Section:	WPAS
Key:	LOCKING
Value:	FALSE



Settings



MOC Settings

General Automatic assignment

☐ Consider operations planned in detail

Personnel display

☒ First name Last name
☐ Last name, first name

Sorting

☒ Personnel no. ☐ Name

OK Cancel

Include detail planned operations

If workforce requirements are generated independent of the detail planned operations, only the workforce requirements are included, which are stored at the workplace for each defined validity periods. No matter whether operations are scheduled or not.

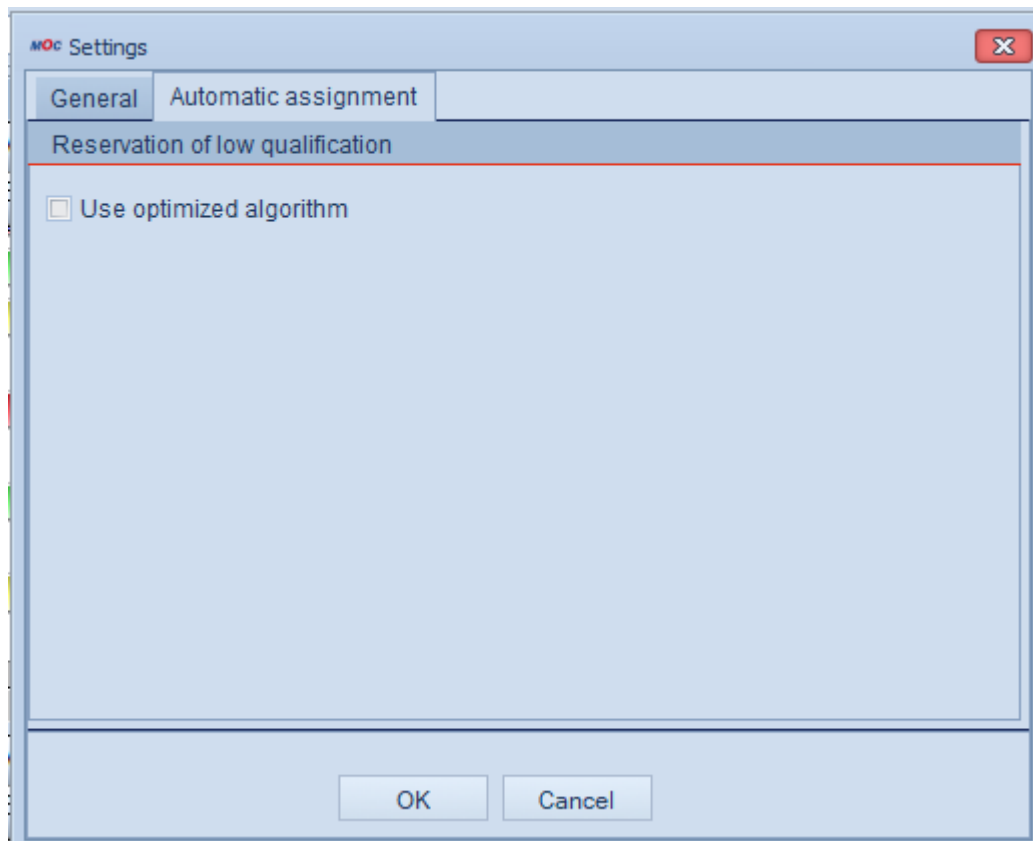
If personnel requirements are generated independent of the detail planned operations, then only periods are included for the workforce requirements that have scheduled operations (=> scheduled operations are privotal). There are three options to store workforce requirements.

First name last name or last name, first name

This field defines whether the last name is shown in front of or behind the first name.

Sorting

This option specifies whether the staff grouped by areas is sorted by the personnel number or by the name for this area.

**Use optimized algorithm**

This option enables the [Optimized, automatic assignment](#).



The configurations are saved with respect to the user per workplace.

**Automatic workplace assignment**

Click this button to call the [Automatic workplace assignment](#).

**Save planning**

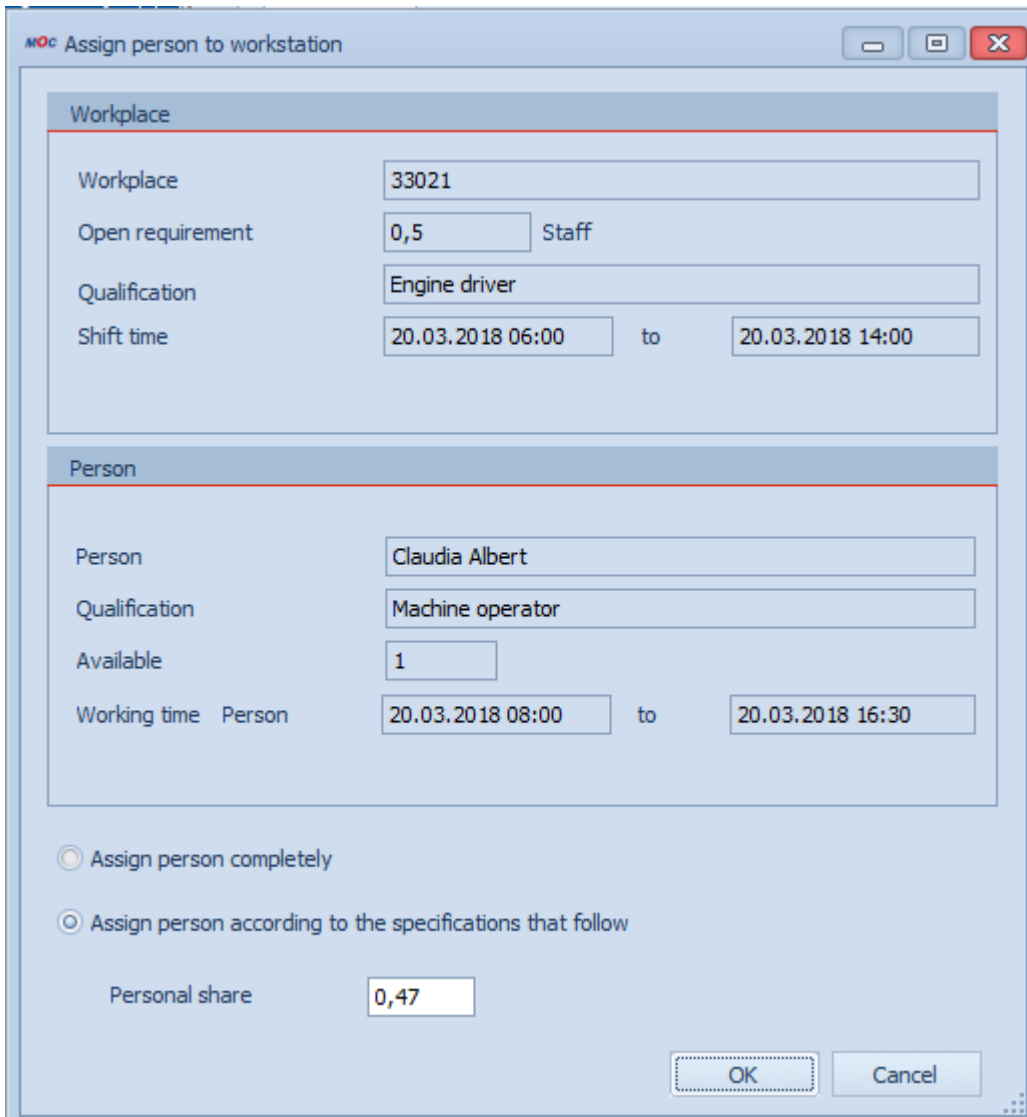
This function saves the current workplace assignment, which is then available to all users.

**Personnel scheduling**

You can call the application *Personnel scheduling* with this button.

Manual assignment

Staff can be planned manually by way of "drag & drop". To do so, select the bar of a person, hold the mouse button down and drag it to the top of a shift for a workplace. The below dialog appears, once you release the mouse button:



Workplace

Workplace: 33021

Open requirement: 0,5 Staff

Qualification: Engine driver

Shift time: 20.03.2018 06:00 to 20.03.2018 14:00

Person

Person: Claudia Albert

Qualification: Machine operator

Available: 1

Working time Person: 20.03.2018 08:00 to 20.03.2018 16:30

☒ Assign person completely

☐ Assign person according to the specifications that follow

Personal share: 0,47

OK Cancel

Workplace

Shows the workplace including the required qualification and the open requirements.

Person

Shows the person with the qualification and availability.

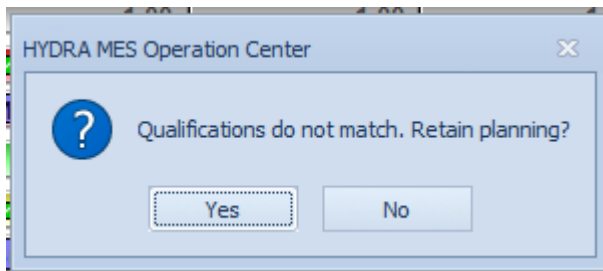
Assign person completely

The person's available time is completely assigned to the workplace, irrespective of whether this exceeds the workplace requirements or not.

Assign person according to the specifications that follow

The missing assignment is compared to the person's availability and the result is shown in this field.
You can change the suggested value.

If the person does not have the required qualification, this message appears:



It is now up to the user to decide whether or not to perform this assignment.

To remove a person from planning, the personnel bar only has to be selected and dragged back from the workplace to the pool of staff. As an alternative, you may also open the context menu by right clicking and selecting the option "cancel assignment".



HYDRA: as of service pack 13

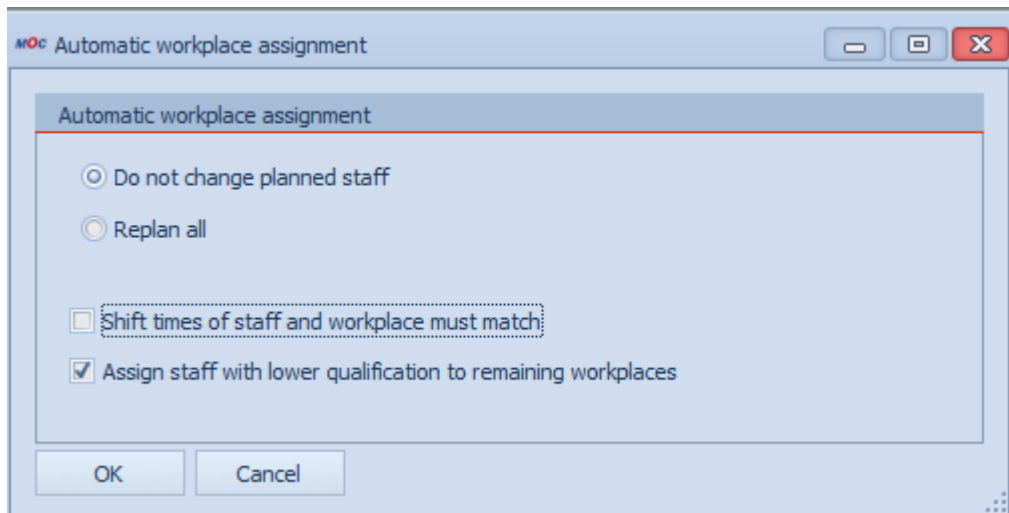
FEDRA: as of version 1.1



You can select a number of people and plan them for a requirement using drag and drop. The dialog mentioned above is called for each planning action.

Automatic assignment

The below dialog appears when starting automatic workplace assignment:



Do not change planned staff, replan all

These options specify whether already existing assignments are to be kept by automatic assignment or whether they are to be removed from planning to be planned once more.

Shift times of staff and workplace must match

Staff is only planned on workplaces with shift times matching the employee's planned standard working time.

Assign staff with lower qualification to remaining workplaces

All staff qualifications are taken into account as part of the assignment. Assignment is performed sorted by the order of priority of the employee's qualifications. If this option is not enabled, only qualifications of the highest order of priority (99) are assigned.

Optimized, automatic assignment

The optimized, automatic assignment function can be enabled by the [settings](#).

In contrast to the automatic assignment function described above that only considers the order of priority for qualifications, the optimized, automatic assignment function also considers the employee's flexibility. This is intended to achieve a higher assignment rate for automatic planning. The optimized assignment function works as follows:

Plan staff with only one assignment option

At first the system checks whether there is staff with only one possible assignment option. This assignment is performed regardless of whether there are employees who have a higher priority for the required qualification. Consequently, this person (with only one assignment option) does not remain unassigned.

Assignment of workplaces with only one possible operator

Then the system checks whether a qualification required by a workplace can only be met by one employee. This employee is assigned to the workplace, irrespective of whether this person still has additional qualifications of a higher order of priority. Consequently, the qualification required by this workplace does not remain unassigned.

Assignment due to the order of priority of qualifications

Just as it is the case with the standard, automatic assignment function, now the employees are assigned to the workplaces sorted by the order of priority of their qualifications. However, after each assignment the two steps described above are carried out again. The reason for it is that since the assignment of an employee can result in persons with only one remaining possible assignment or workplaces with only one possible operator being available.



The optimized algorithm is only used if the option "Assign staff with lower qualification to remaining workplaces" is enabled for automatic assignment.

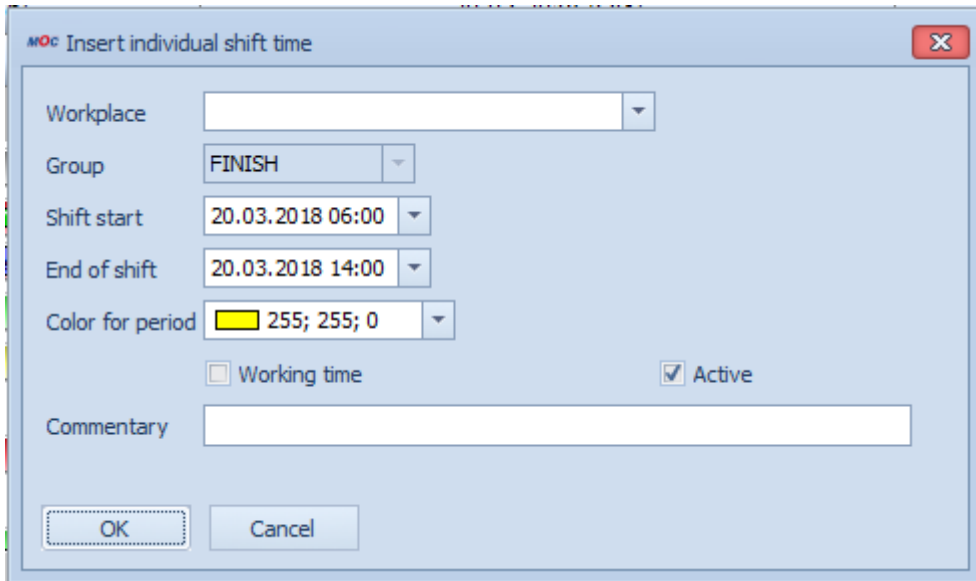
Replan assignments

If you want to change planning for an employee assigned to a workplace and replan this employee to another workplace, you can do it by "drag & drop" as it is the case with manual assignments. Once you release the mouse button, the same dialog occurs as with manual assignments.

Individual shift times

Using [Individual shift times](#), you can define additional working times and times off deviating from the planned shift model for individual workplaces.

In order to change a complete or parts of the shift and to change it into "time off", just right-click the relevant shift and select the entry "Insert individual shift time" from the context menu. The following dialog opens where you can enter the period for the time off and a comment:



MOG Insert individual shift time

Workplace

Group

Shift start

End of shift

Color for period

☐ Working time ☒ Active

Commentary

OK Cancel

In order to add additional working time, just right-click a time off and select the entry "Insert individual shift time" from the context menu. Then hold the left mouse button down and drag the period for which you want to add the additional working time. Once you have released the mouse button, the above-mentioned dialog opens where you can correct the time and enter a comment. Here, you must enable the option "working time" in order to specify that it is planned working time and no time off.

Individual shift times can be changed and deleted in the application [Individual shift times](#).

4 Individual shift/assignment times

Overview

Menu	Production control → Preparations for production → Individual shift/assignment times
Transaction code	mdistmf
Function authorization	mdistmf.*

Purpose

You can define individual shift/assignment times within the [Graphic planning](#) and the [Workplace assignment](#) and specify for a workplace within a specific period of time, whether this time is to be considered as working time or idle time.

This enables short-term modifications with respect to the availability of workplaces, without having to change the planned shift model.

This application manages times without shift or shift times that have been assigned to one or several workplaces within the functions Graphic planning or Workplace assignment.

Integration

Normally, individual shift/assignment times are directly defined by the corresponding functionality provided in the Graphic planning or the Workplace assignment.



These individual shift/assignment times do not affect collection and posting within the scope of shop floor data collection.

Requirements

The product group Graphic planning or Workplace assignment is in use.

Selection criteria

The application provides the following selection criteria:

Group

You can restrict the entries to a specific group by using this combo box.

Workplace

Enter a workplace to view only entries for this workplace.

Period from

Optionally, you can use this input field to enter the beginning of a period as of which you want to display the entries in the application.

Working time

This option selects

- ☐ times without shift only
- ☒ working times only
- ☐ both times



Please note that 3 states are available with this checkbox. If you require data that is missing, have a look at the checkbox setting. We recommend to set the checkbox to the option ☐.

Active

You can enable/disable individual shift/assignment times. The Graphic planning does not include disabled entries.



Please note that 3 states are available with this checkbox. If you require data that is missing, have a look at the checkbox setting. We recommend to set the checkbox to the option ☐.

Field descriptions

Workplace

Workplace for which the entry (individual shift/assignment time) has been created.

Group

Group of the workplace for which the entry (individual shift/assignment time) has been created.

Period from

Point in time when the individual shift/assignment time begins.

Period until

Point in time when the individual shift/assignment time ends.

Working time

This option describes, if it is

- ☐ a time without shift
- ☒ working time.

If it is not working time, but a time without shift, the workplace capacity is not available during this period. Therefore, you cannot plan an operation.

Active

You can enable/disable individual shift/assignment times. The Graphic planning does not include disabled entries.

We recommend to set this option in general to .

Comment

You can store a short comment for this individual shift time including further details.



In the Graphic planning, you can additionally specify a color for the time without shift when defining an individual shift/assignment time.

In the Graphic planning, you can display the comment that is stored for an individual shift in the tooltip of the corresponding individual shift (as of HLS 8.2). Enable (disable) the tooltip using an INI data entry.

- Name: HLS
- Section: SCHEDULING
- Key: DISPLAY_TOOLTIP_FOR_ISTMF
- Value: J
- Active: [selected]

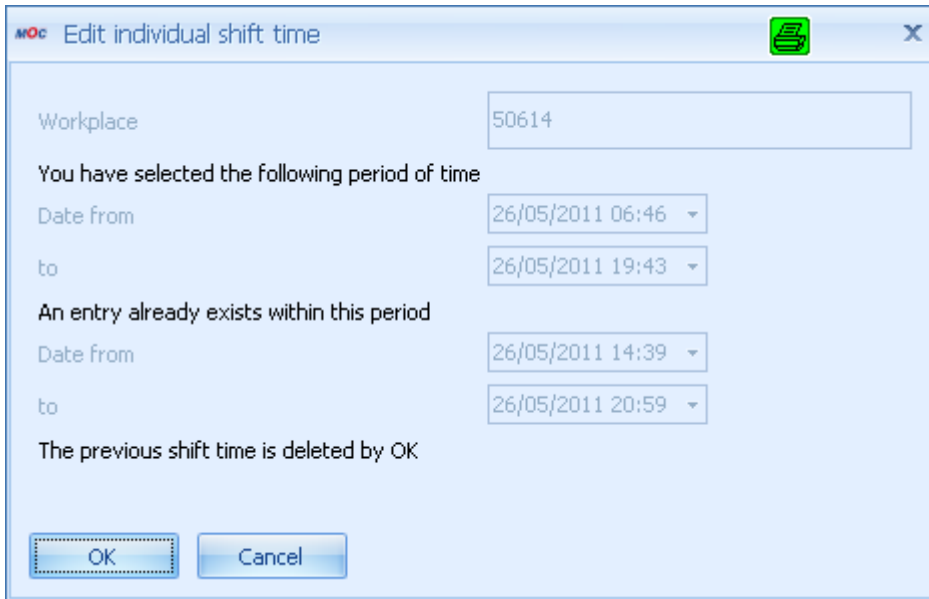


If you enable the tooltip display, the presentation color of times without shift changes in the Graphic planning. In addition, the workplace bars are displayed one level before the times without shift.

Editing functions

Use the available buttons to create or edit individual shift/assignment times.

Only one entry can exist at a workplace for each period. If you make an entry and the system detects that an entry already exists for this period, you can either delete the previous entry or cancel this entry.



If multiple individual shift times exist, this dialog is opened for each existing shift time, and the user can decide whether or not to delete the entries.



Please be careful not to define overlapping periods for one workplace.

These additional shift times are only used in Shop Floor Scheduling, not as part of data collection.

The user can only change, view and delete machines that belong to the responsibility area he/she is authorized for.

5 Workplace Schedule

Overview

Menu	Production control -> Workforce requirements planning -> Workplace schedule Human resources management -> Workforce requirements planning -> Workplace schedule
Transaction code	wpsch
Function authorization	wpsch

The application "Workplace schedule" shows equivalent to the [Workforce requirements plan](#) the result of the workplace assignment. The system displays the "Workplace schedule" as a graphic in form of a report whereas the "Workforce requirements plan" is displayed as a table. Another difference is that the "Workforce requirement plan" shows the assignment from the point of view of the employee (when do I work and where / when am I off work) and the "Workplace schedule" relates to the workplace (who works in which shift and the relevant qualifications).

The screenshot shows the 'Workplace schedule' application window. It includes a menu bar with 'Main page' and 'Profile', and a toolbar with icons for 'Request data', 'Cancel', 'Save', and 'Help'. The 'Selection' section shows a 'Planning profile' dropdown and a date range from '03.03.2015' to '04.03.2015'. The main content area displays a report titled 'Workplace schedule' for the date '03.03.2015'. The report is organized into sections: 'FINISH Finishing' and 'INJ.MOLD1 InjectionMolding Machines'. Each section contains a table with columns for time slots (06:00 - 14:00, 14:00 - 22:00, 22:00 - 06:00) and rows for different roles (Engine driver, Machine operator, Fork-lift driver) and employees (Meyers, Harald Edward, Meyers, Joseph, Pernikova, Lisa, Baker, Jake, Green, Mary, Berry, Claire, White, Susan, Smith, Christian).

FINISH Finishing		03.03.2015		
		06:00 - 14:00	14:00 - 22:00	22:00 - 06:00
33021 Manual Workplace Deburring, Cleaning	Engine driver		Meyers, Harald Edward	Meyers, Joseph
	Machine operator		Pernikova, Lisa	
	Fork-lift driver	Baker, Jake		Smith, Christian
33022 Manual Workplace Deburring, Cleaning	Engine driver		Meyers, Harald Edward	Meyers, Joseph
	Machine operator	Green, Mary	Berry, Claire	White, Susan
	Fork-lift driver	Baker, Jake		Smith, Christian

INJ.MOLD1 InjectionMolding Machines		03.03.2015		
		08:00 - 14:00	14:00 - 16:30	17:00 - 20:00
50610 Engel	Fitter		Miller, Steven	

The display of the workplace assignment is grouped by workplace group and date. When the system displays workplaces and the assigned qualifications it is within such a constellation. On the right, you can find the shift times and the personnel planned per shift.

Selection criteria

The application provides the following selection criteria:

Planning profile

You can select a planning profile to restrict the workplace schedule displayed.

Date from, to

You can select a period to display the workplace schedule.

6 Workforce Requirements Plan

Summary

Menu	Production Control → Personnel Scheduling → Workforce Requirements Plan Human Resource Management → Personnel Scheduling → Workforce Requirements Plan
Transaction code	pesch
Function authorization	pesch

The Workforce Requirements Plan display shows which person is assigned to which workplace and if there are still any unassigned, free times among the employees.

Person	Name	Planning	Absence	Weekday	Date	Time	End of work...	Workplace	Designation
667	Smith, Christian	FLX		Tuesday	3/5/2013	8:00	16:30		
40256	Pernikova, Lisa	S	HLD	Tuesday	3/5/2013	14:00	22:00		
40563	Green, Mary	S		Tuesday	3/5/2013	14:00	22:00	33021	Manual Workplace Debur...
40789	Sheen, Edward	N		Tuesday	3/5/2013	22:00	6:00	33021	Manual Workplace Debur...
50014	Albert, Claudia			Tuesday	3/5/2013	8:00	14:00	33021	Manual Workplace Debur...
50014	Albert, Claudia			Tuesday	3/5/2013	14:00	16:30	33021	Manual Workplace Debur...
50201	Mayer, Hugo Egon	N		Tuesday	3/5/2013	22:00	6:00		
59881	Brown, Albert			Tuesday	3/5/2013	8:00	16:30		
85741	Meyers, Joseph	F		Tuesday	3/5/2013	6:00	14:00	33021	Manual Workplace Debur...
85741	Meyers, Joseph	F		Tuesday	3/5/2013	6:00	14:00	33022	Manual Workplace Debur...
93856	Miller, Steven			Tuesday	3/5/2013	8:00	16:30		
96665	Hanson, Betty			Tuesday	3/5/2013	8:00	14:00	52276A	Paint Line 1a
96665	Hanson, Betty			Tuesday	3/5/2013	14:00	16:30	52276A	Paint Line 1a
10002541	Berry, Claire	N		Tuesday	3/5/2013	22:00	6:00	33022	Manual Workplace Debur...

Selection criteria

The following selection criteria are available in the application:

Person from/ to, company, area, cost center

Selection of the people to be displayed

Date from/ to

Period for which the personnel schedule is to be displayed.

Workplace

Machine/ workplace

Planning profile

The display of the personnel schedule can be limited using a planning profile. This field can only be accessed if the additional function Advanced selection and visualization is present.

Summarize absences

Contiguous days with absences for a person are output in a row with the start and end date.

Display days without normal time

The list also includes days off on which a normal time is not stored in the working time day type of HYDRA-PZE. In this way, in a longer evaluation period one calendar view per person is possible.

If the personnel schedule was limited using a planning profile, only those days are displayed on which the employees have an assignment for a workplace of the corresponding planning profile.



The value in the *Free time* column contains only the assignments within this planning profile and for this reason it is only correct if the employee was not assigned outside of the planning profile.